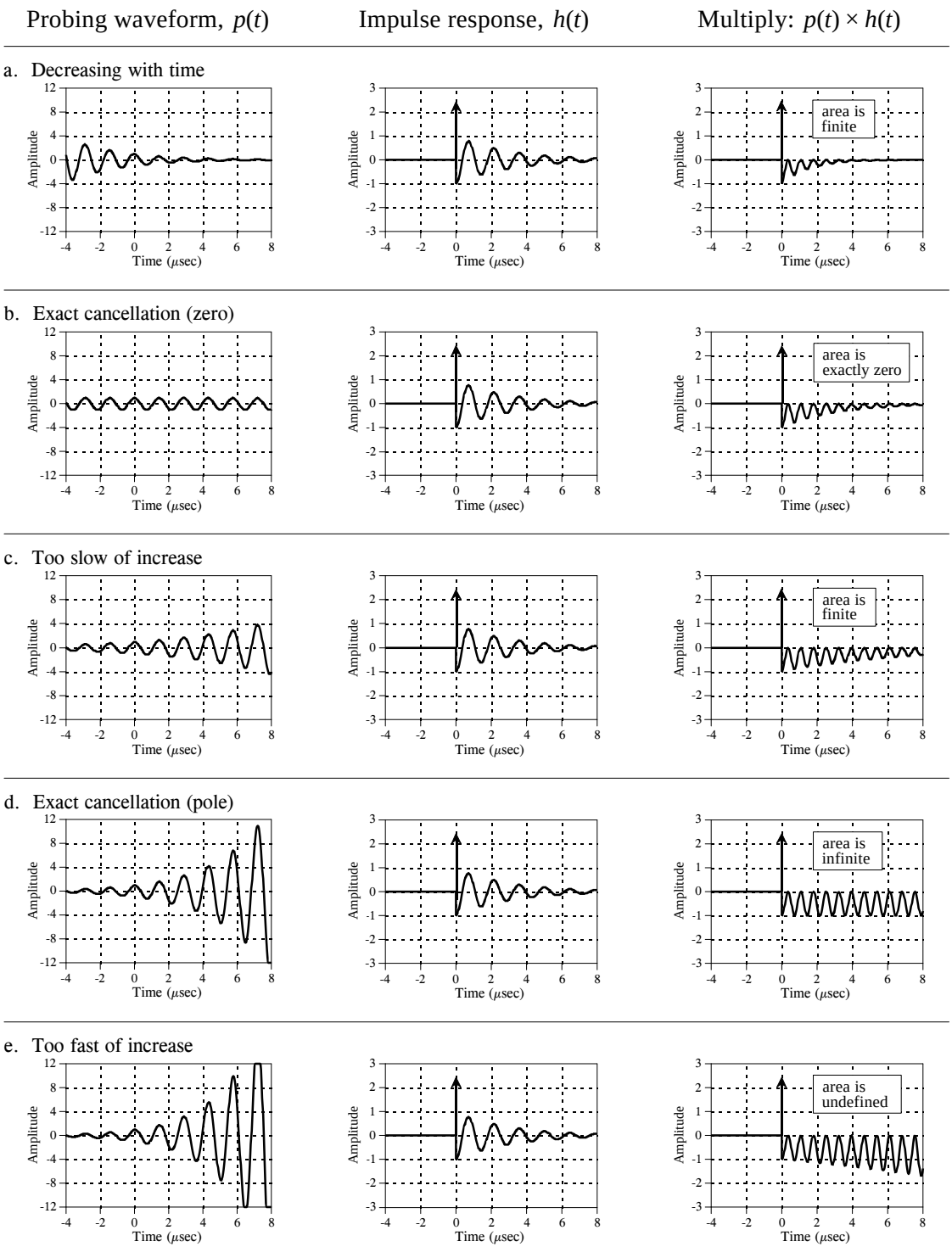


FIGURE 32-5
Probing the impulse response. The Laplace transform can be viewed as probing the system's impulse response with various exponentially decaying sinusoids. Probing waveforms that produce a cancellation are called *poles* and *zeros*. This illustration shows five probing waveforms (left column) being applied to the impulse response of a notch filter (center column). The locations in the s-plane that correspond to these five waveforms are shown in Fig. 32-4.